Sensitivity of Log-Derived Water Saturation to Archie Parameter Uncertainty: A Pickett-Plot Analysis and Volumetric Propagation

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Abstract

Water saturation derived from wireline logs is not a measurement but an inference through Archie's equation, whose cementation exponent m, saturation exponent n, and tortuosity factor a are empirical parameters fitted to core. This paper reviews the physical basis of the Archie model, the boundaries of its validity in shaly and heterogeneous formations, and the propagation of parameter uncertainty into computed saturation and original-oil-in-place. Using a Pickett-plot construction we show that a plausible uncertainty in the cementation exponent maps to a large uncertainty in saturation near economic cutoffs, and that saturation uncertainty, combined with porosity and net-pay uncertainty, dominates the volumetric estimate. We argue that formation-evaluation deliverables should report saturation as a distribution traceable to a stated calibration rather than as a single deterministic curve. No new laboratory data are reported; the analysis uses published parameter ranges.

Keywords— petrophysics, Archie equation, water saturation, Pickett plot, cementation exponent, uncertainty propagation, volumetrics, SPE-PRMS.

1. Introduction

Formation evaluation converts indirect physical measurements into the parameters that govern hydrocarbon volume and flow. Wireline and logging-while-drilling tools respond to bulk properties—natural gamma radiation, electrical resistivity, bulk density, hydrogen index, and acoustic slowness—none of which is porosity or saturation. Each becomes a reservoir property only through a petrophysical model calibrated against core. The dominant model for water saturation remains the relation published by Archie in 1942 [1], which ties saturation to formation resistivity through porosity and two empirical exponents.

Archie's relation was derived from clean, water-wet sandstones with no conductive clay. Applied outside that envelope—most consequentially in shaly sands and heterogeneous carbonates—it systematically overestimates water saturation, because clay-bound water provides a parallel conduction path the model omits. The shaly-sand models of Waxman and Smits [2] and the Dual-Water formulation of Clavier and colleagues [3] were developed to correct this, and the choice between them is a material engineering decision. The purpose of this paper is to make the sensitivity explicit: to show how uncertainty in the Archie parameters propagates into saturation and, through the volumetric equation, into reserves.

2. The Archie Model and Its Validity Envelope

Archie's equation states that water saturation raised to the saturation exponent equals the product of the tortuosity factor and formation-water resistivity, divided by the product of porosity raised to the cementation exponent and true formation resistivity. Its physical premise is that the rock matrix is an insulator and that all conduction occurs through connected brine in the pore network. That premise is the model's entire validity envelope [1], [4].

The cementation exponent m is near 2.0 in clean consolidated sandstone but ranges from roughly 1.3 in unconsolidated sand to well above 2.5 in vuggy or moldic carbonate, where isolated pore space raises resistivity without contributing a conductive path. The saturation exponent n is near 2.0 in strongly water-wet rock but rises substantially in oil-wet systems, where the conductive water phase loses continuity; wettability is therefore an electrical parameter, not only a capillary one [4], [5]. The tortuosity factor a and the exponent m trade off against one another in the fit, so quoting one without the other is meaningless: they are a matched pair from a single regression.

Table I. Archie parameter ranges by rock type

Rock type	Cementation m	Saturation n	Controlling factor
Unconsolidated sand	~1.3–1.6	~1.8–2.0	Grain-contact geometry; minimal cement; well-connected pores
Consolidated clean sandstone	~1.8–2.1	~1.8–2.2	Degree of cementation; Archie's original calibration domain
Vuggy / moldic carbonate	~2.2–3.0+	Highly variable	Isolated vugs raise resistivity without conductive path
Fractured carbonate	<1.5 possible	Variable	Fractures short-circuit conduction; m loses physical meaning

Rock type	Cementation m	Saturation n	Controlling factor
Oil-wet rock (any lithology)	Model-dependent	Can exceed 3–4	Discontinuous water film; conduction path loses connectivity

3. Pickett-Plot Sensitivity Analysis

The Pickett plot is a graphical solution of Archie's equation obtained by cross-plotting true resistivity against porosity on logarithmic axes [6]. Because the equation is a power law in both variables, lines of constant water saturation appear as straight lines whose slope is the negative of the cementation exponent, and whose intercept encodes the tortuosity factor and formation-water resistivity. The construction is valuable precisely because it exposes the parameters visually: the analyst reads m from the slope of the 100%-saturation trend through the water-bearing points, rather than assuming it.

Figure 1 shows the construction for a clean-sand parameter set. The clustering of log data along the fabric of constant-saturation lines is the qualitative test of parameter validity: systematic curvature or scatter indicates that a single global m does not describe the interval, which is the signature of mixed rock types or clay conductivity. The sensitivity that concerns reserves is visible directly. Because the constant-saturation lines fan out from the water line, a fixed uncertainty in the slope m produces a saturation uncertainty that grows as porosity falls, so the poorest reservoir rock—precisely the rock near economic cutoff—carries the largest saturation uncertainty.

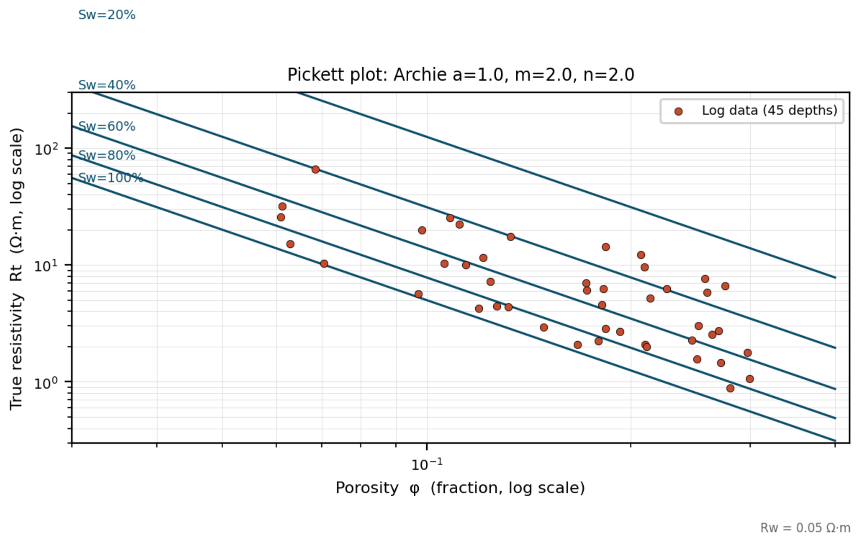

Figure 1. Pickett plot for Archie parameters a = 1.0, m = 2.0, n = 2.0 with formation-water resistivity Rw = 0.05 Ω·m. Straight lines are loci of constant water saturation; their common slope is –m. Points are synthetic log readings at 45 depths with lognormal measurement scatter. The widening vertical separation of the saturation lines at low porosity is the source of elevated saturation uncertainty near reservoir cutoff.

A quantitative reading follows from the slope sensitivity. Taking the clean-sand exponent as 2.0 with an uncertainty of plus or minus 0.2, a range well within the spread reported across special-core-analysis studies [4], [5], the implied saturation at a mid-range porosity shifts by several saturation units, and the shift roughly doubles at the low-porosity end of the pay interval. Where the cutoff decision itself depends on saturation, this converts parameter uncertainty directly into uncertainty about whether a given foot is counted as pay.

4. Propagation into Volumetrics

Original oil in place is the product of bulk rock volume, net-to-gross, porosity, and hydrocarbon saturation, divided by the oil formation volume factor. Because the terms multiply, relative uncertainties combine approximately in quadrature rather than additively. A 15% relative uncertainty in net pay, 10% in porosity, and 20% in hydrocarbon saturation yields roughly 27% uncertainty in oil in place before any consideration of recovery factor, which is itself the most uncertain term in the chain [7], [8].

Two consequences follow. First, the base case is not the mean: the product of positively distributed terms is approximately lognormal and therefore right-skewed, so a deterministic mid-value systematically misstates the expectation. Second, the formation volume factor, frequently treated as a known constant, is a pressure-dependent PVT property; using a single value across a reservoir with significant pressure variation introduces a systematic bias that no statistical treatment of the other terms will surface [8]. These are the arguments for probabilistic volumetrics with stated input distributions.

5. Reserves Reporting under SPE-PRMS

The Petroleum Resources Management System classifies recoverable volumes by commercial maturity and uncertainty [9]. The probabilistic categories are specific: proved (1P) corresponds to at least a 90% probability that the quantity recovered equals or exceeds the estimate, proved-plus-probable (2P) to at least 50%, and proved-plus-probable-plus-possible (3P) to at least 10%. The categories are cumulative and their probabilities are not additive. Reporting a single reserves figure without its category basis discards precisely the uncertainty information the classification exists to convey, and the saturation sensitivity analysed above is one of the inputs that populates the 1P–3P spread.

6. Discussion and Recommendations

The practical implication is procedural. Archie's exponents should be fitted to field-specific special-core-analysis data and reported as a matched set of a, m, and n with the depth interval and sample count over which they were regressed; an uncalibrated Archie curve is a hypothesis presented as a measurement. Where cation-exchange capacity is non-trivial, a shaly-sand model should be run and the saturation difference against Archie reported as an explicit sensitivity rather than absorbed silently. Volumetrics should be reported probabilistically, and every reserves number should carry its SPE-PRMS category into every downstream document, including summaries prepared for non-technical readers.

7. Conclusion

Log-derived water saturation inherits the uncertainty of the Archie parameters, and that uncertainty is largest in the poorest reservoir rock, where the pay decision is most marginal. Propagated through the volumetric equation, saturation uncertainty is a leading contributor to the reserves range. Treating the Archie parameters as calibrated quantities with stated uncertainty, rather than as constants, is the low-cost step that keeps a reservoir model defensible.

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